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Clinical Phenotypes and Long-term Management of Inflammatory Bowel Disease in a Specialized Private Unit: A 12-year Retrospective Cohort Study

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ABSTRACT

Introduction: Real-world data from specialized private Inflammatory Bowel Disease (IBD) units are essential for a comprehensive understanding of disease patterns. This study describes the clinical characteristics and therapeutic management of a cohort followed in a specialized private IBD unit over 12 years, with a focus on disease severity and the use of advanced therapies.

Material and methods: We retrospectively analyzed 244 adult patients, 129 Ulcerative Colitis (UC), 99 Crohn's Disease (CD), and 16 Inflammatory Bowel Disease-Unclassified (IBD-U) followed at a private IBD unit in Athens, Greece, between 2014 and 2025. Clinical phenotypes were classified according to the Montreal criteria. Primary outcomes included the rates of biological and small molecule therapy, and the necessity for surgery.

Results: Complicated disease phenotypes were prevalent, with 52.5% of CD patients presenting with stricturing or penetrating behavior (Montreal B2/B3) and 58.9% of UC patients having extensive colitis (E3). Cumulative exposure to advanced therapies (biologics and small molecules) was 41.0%. IBD-related surgery rates were 32.3% for CD and 8.5% for UC.

Conclusions: These findings underline the clinical complexity and therapeutic patterns observed in this single private cohort. However, they should be interpreted within the limitations of a retrospective, single-center design without a public comparator.

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Fenotipos clínicos y manejo a largo plazo de la enfermedad inflamatoria intestinal en una unidad privada especializada: estudio de cohortes retrospectivo de 12 años

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RESUMEN

Introducción: Los datos de práctica clínica real procedentes de unidades privadas especializadas en Enfermedad Inflamatoria Intestinal (EII) son esenciales para una comprensión integral de los patrones de la enfermedad. Este estudio describe las características clínicas y el manejo terapéutico de una cohorte seguida en una unidad privada especializada en EII durante 12 años, con un enfoque en la gravedad de la enfermedad y el uso de terapias avanzadas.

Material y métodos: Analizamos retrospectivamente a 244 pacientes adultos, 129 con Colitis Ulcerosa (CU), 99 con Enfermedad de Crohn (EC) y 16 con Enfermedad Inflamatoria Intestinal No Clasificada (EII-U), seguidos en una unidad privada de EII en Atenas, Grecia, entre 2014 y 2025. Los fenotipos clínicos se clasificaron según los criterios de Montreal. Los resultados primarios incluyeron las tasas de terapia biológica y con moléculas pequeñas, y la necesidad de cirugía.

Resultados: Los fenotipos de enfermedad complicada fueron prevalentes, con un 52,5% de los pacientes con EC que presentaban comportamiento estenosante o penetrante (Montreal B2/B3) y un 58,9% de los pacientes con CU que presentaban colitis extensa (E3). La exposición acumulada a terapias avanzadas (biológicos y moléculas pequeñas) fue del 41,0%. Las tasas de cirugía relacionada con la EII fueron del 32,3% para la EC y del 8,5% para la CU.

Conclusiones: Estos hallazgos subrayan la complejidad clínica y los patrones terapéuticos observados en esta cohorte privada única. Sin embargo, deben interpretarse dentro de las limitaciones de un diseño retrospectivo y unicéntrico sin un comparador público.

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1. INTRODUCTION

The global burden of Inflammatory Bowel Disease (IBD) reflects a complex interplay of sociodemographic indices, age, and environmental factors. As of recent epidemiological assessments, over 3.8 million cases of IBD have been reported globally, demonstrating significant disparities in incidence and clinical patterns across geographic regions [1-3]. While IBD incidence appears to be stabilizing or slightly declining in high-income nations, it continues to rise sharply in newly industrialized and low-to-middle-income regions. This global evolution contributes to an increasing overall disease burden in terms of disability-adjusted life years (DALYs), posing substantial economic and management challenges to healthcare systems worldwide [4].

Concurrently, the demographic profile of IBD patients is undergoing a notable shift. Recent data highlight a disproportionate increase in disease incidence among female patients and the elderly population [5, 6]. Managing IBD in older adults introduces unique clinical complexities, including age-related comorbidities, higher vulnerability to treatment-related adverse events, and specific geriatric healthcare needs. Furthermore, regional and ethnic variations significantly influence disease phenotype, extraintestinal manifestations, and clinical outcomes [7-9], emphasizing the necessity for tailored healthcare policies and personalized treatment protocols [10].

To date, most epidemiological and clinical outcome data have originated from academic institutions or public national registries. In Greece, published information regarding IBD patterns is similarly derived almost exclusively from public tertiary centers, leaving a significant gap in knowledge regarding cohorts managed in specialized private settings. However, private hospitals and specialized community-based IBD units are increasingly becoming crucial providers of comprehensive care, particularly in healthcare systems where a substantial proportion of patients seek private medical management. Patients attending specialized private tertiary units may present with

different socioeconomic profiles, distinct health-seeking behaviors, or different utilization patterns of advanced therapies. Evaluating data from high-volume, specialized private centers is therefore essential to capture the full spectrum of real-world IBD care and benchmark outcomes against national and international standards.

In this context, the present study aims to provide a detailed clinical characterization of a 12-year patient cohort followed at a specialized private tertiary IBD unit in Greece. Specifically, we evaluated disease phenotypes, long-term therapeutic trajectories, advanced therapy utilization, and surgical outcomes. These findings offer contemporary insights into the clinical complexity managed within the Greek private healthcare sector and highlight the strategic role of dedicated private IBD infrastructure.

2. MATERIAL AND METHODS

2.1. STUDY DESIGN AND PATIENT COHORT

This was a single-center; retrospective, descriptive study conducted at the IBD Unit of the Private "*Metropolitan General" Hospital,* Holargos, Greece. The unit provides specialized outpatient and inpatient services for patients with IBD and serves as a referral center. We used data from an anonymized database of adult (>18 years old) IBD patients from 2014 to 2025. The initial database contained 278 patient records. After stringent data cleaning and exclusion of entries with missing primary diagnostic data, a final cohort of 244 patients with confirmed IBD was included for analysis. In more details, patients excluded were those under 18 years old, patients with incomplete clinical/diagnostic medical records, and patients with non-IBD colitis (e.g., infectious, ischemic, or microscopic colitis). The patients were stratified into three groups based on their definitive diagnosis: Ulcerative Colitis (UC), Crohn's Disease (CD), and IBD Unclassified (IBD-U).

2.2. DATA COLLECTION AND DEFINITIONS

Data were extracted from the electronic patient files and the accompanying clinical database. The following parameters were collected:

- Demographics: Sex, current age, hospitalization rates.
- Disease Characteristics: Disease type (UC, CD, IBD-U), age at diagnosis, disease duration, number of hospitalizations, severity (categorized as 1: Mild, 2: Moderate, 3: Severe), and disease location/phenotype according to the Montreal classification. In more details CD patients were characterized by age at diagnosis (A1: <16, A2: 17–40, A3: >40 years), location (L1: ileal, L2: colonic, L3: ileocolonic, L4: upper GI), and behavior (B1: non-stricturing/non-penetrating, B2: stricturing, B3: penetrating, p: perianal modifier). UC extent was categorized as E1 (proctitis), E2 (left-sided), and E3 (extensive).
- Comorbidities and Lifestyle: Presence of autoimmune diseases (AID), family history of IBD or AID, smoking status (1: Current smoker, 2: Never/Past smoker), use of Non-Steroidal Anti-inflammatory Drugs (NSAIDs), and history of appendectomy.
- Extraintestinal Manifestations (EIMs). Systematically recorded EIMs included peripheral/axial arthralgia or arthritis, cutaneous lesions (erythema nodosum, pyoderma gangrenosum), and ocular involvement (iridocyclitis/uveitis).

Treatment definitions and History: Cumulative exposure to Mesalazine, Corticosteroids, Immunomodulators and Biologic Agents. Related Surgery (colectomy, proctocolectomy, ileal pouch-anal anastomosis, stricturoplasty, abscess drainage).

2.3. STATISTICAL ANALYSIS

Categorical variables are presented as frequencies and percentages and were compared using the Pearson Chi-Square (chi2) test or Fisher's exact test. Continuous variables are presented as mean \pm standard deviation (SD) and were compared using One-Way Analysis of Variance (ANOVA) or non-parametric Kruskal-Wallis tests. A two-tailed p-value <0.05 was considered statistically significant. Statistical analyses were performed using SPSS version 28.0. Due to the inherent selection bias of a private unit and the retrospective nature of the study, the focus remained on descriptive clinical patterns rather than comparative inference against external populations.

2.4. ETHICS

The study was conducted under our institutional ethics committee approval ("Metropolitan General" Hospital,

Number of approval: 195, March 26, 2025).

3. RESULTS

3.1. DEMOGRAPHICS AND BASELINE CHARACTERISTICS

A total of 244 IBD patients were included in the final analysis. The cohort was comprised of 129 patients with UC (52.9%), 99 with CD (40.6%), and 16 with IBD-U (6.6%). Table 1 shows the demographic and baseline clinical characteristics stratified by disease subtype. No statistically significant difference in sex distribution was found across the three IBD types (p=0.448) (Table 1). The cohort's overall mean age was 40.5 years. There was a significant difference in mean age among the groups (p=0.041), with UC patients, on average, older than CD patients. Regarding age at diagnosis, a significant difference was observed (p<0.001). CD patients were diagnosed approximately 8.8 years earlier than UC patients (30.1 vs. 38.9 years). CD patients exhibited a significantly longer disease duration (8.1 years) compared to UC patients (3.7 years) (p<0.001) (Table 2). However, this could reflect a potential referral bias, as CD patients with longer disease histories and more complex disease are managed in this specialized private unit. Regarding smoking habits in the whole cohort of patients, smokers/former smokers (Yes/Ex-Smoker) were compared with non-smokers (No). The percentage of active smokers was three times higher in CD (28.3%) than in UC (7.8%), with the difference statistically significant (p<0.001). Overall, a significantly higher proportion of CD patients (40.4%) had a history of smoking (active or former) compared to UC patients (21.7%) (p=0.004).

3.2. DISEASE PHENOTYPE (MONTREAL CLASSIFICATION)

The majority of UC patients in the cohort (58.9%, 76/129) have Extensive Colitis (E3). A smaller proportion (31.8%, 41/129) had Left-sided Colitis (E2), and an even smaller proportion (9.3%, 12/129) had proctitis.

Regarding CD, almost one in two (45.5%, 45/99) had ileocolonic disease (L3), one/third had ileal (L1) (35.4%, 35/99), one/fifth had large bowel (L2) involvement (15.2%, 15/99), and a small proportion (4.0%, 4/99) had upper GI involvement.

Table 1: Demographic and Baseline Characteristics Stratified by IBD Subtype

Characteristic	Cohort (N=244)	UC (n=129)	CD (n=99)	IBD-U (n=16)	p-value
Male	136 (55.7%)	76 (58.9%)	51 (51.5%)	9 (56.3%)	0.448
Female	108 (44.3%)	53 (41.1%)	48 (48.5%)	7 (43.7%)	
Age Mean ± SD (years)	40.5±14.8	42.6±14.2	38.2±15.2	39.5±14.7	0.041
Age at Diagnosis Mean ± SD (years)	35.1±14.2	38.9±13.5	30.1±13.9	33.4±11.6	<0.001*
Disease Duration Mean ± SD (years)	5.4±7.1	3.7±5.1	8.1±8.6	6.1±7.3	<0.001*
Current Smoking n (%)	40/244	10 (7.8%)	28 (28.3%)	2 (12.5%)	<0.001*
Ex-Smoker n (%)	33/244	18 (14.0%)	12 (12.1%)	3 (18.8%)	0.821
Never Smoking n (%)	171/244	101 (78.3%)	59 (59.6%)	11 (68.8%)	0.004

Non-stricturing-non-penetrating CD (B1) accounted for the majority of patients (47.5%, 47/99). Stricturing disease (B2) was noticed in 28.3% (28/99) of patients, and penetrating/fistulizing (B3) in 24.2% (24/99).

Table 2: Results according to Montreal classification			
Disease	**Phenotype**	**Code**	**Number of patients, n (%)**
Crohn's Disease (N=99)	**Age (A)**	**A1:** < 17 years **A2:** 17–40 years **A3:** > 40 years	*Not systematically subgrouped in base database* Mean age at diagnosis: 30.1 ± 13.9 years
	Location (L)	**L1:** terminal ileum **L2:** colon **L3:** ileocolon **L4:** upper gastrointestinal	35 (35.4%) 15 (15.2%) 45 (45.5%) 4 (4.0%)
	Behavior (B)	**B1:** non-stricturing, non-penetrating **B2:** stricturing **B3:** penetrating **p:** perianal modifier	47 (47.5%) 28 (28.3%) 24 (24.2%) 26 (26.3%)
Ulcerative Colitis (N=129)	**Extent (E)**	**E1:** ulcerative proctitis **E2:** left-sided ulcerative colitis **E3:** extensive ulcerative colitis	12 (9.3%) 41 (31.8%) 76 (58.9%)
	Severity (S)	**S0:** clinical remission **S1:** mild UC **S2:** moderate UC **S3:** severe UC	*Retrospective total severity scoring:* Baseline disease severity evaluated categorically in Section 2.2

The majority of patients with CD present with complicated disease (B2 stricturing /B3 penetrating phenotype- 52.5%). A significant proportion of patients also exhibited perianal disease (p) (26.3%). A dedicated summary Table 2 shows the results according to Montreal classification.

Table 3: Clinical Burden and Complications					
EIMs/Complications	**Total (n=244)**	**UC (n=129)**	**CD (n=99)**	**IBD-U (n=16)**	**p-value**
Any EIM, n (%)	64 (26.2%)	22 (17.1%)	39 (39.4%)	3 (18.8%)	<0.001*
Arthralgia/Arthritis	45 (18.4%)	14 (10.9%)	28 (28.3%)	3 (18.8%)	<0.001*
Erythema Nodosum	12 (4.9%)	2 (1.6%)	10 (10.1%)	0 (0.0%)	0.011*
Iridocyclitis/Uveitis	4 (1.6%)	0 (0.0%)	4 (4.0%)	0 (0.0%)	0.068
Strictures	31 (12.7%)	3 (2.3%)	28 (28.3%)	0 (0.0%)	<0.001*
Abscesses	21 (8.6%)	1 (0.8%)	20 (20.2%)	0 (0.0%)	<0.001*
Fistulas	18 (7.4%)	0 (0.0%)	18 (18.2%)	0(0.0%)	<0.001*
Perianal Disease	26 (10.6%)	0 (0.0%)	26 (26.3%)	0 (0.0%)	<0.001*

3.3. EXTRAINTESTINAL MANIFESTATIONS (EIMS AND COMPLICATIONS)

The presence of cumulative EIMs was assessed across the cohort. The overall prevalence of EIMs was 26.2% (n=64) (Table 3). The prevalence of any EIM was significantly higher in CD patients (39.4%) compared to UC patients (17.1%) ($p < 0.001$). Both Arthralgia/Arthritis and Erythema Nodosum were significantly more prevalent in the CD group compared to the UC group. The frequency of IBD-related complications showed significant differences. Structural complications (stenosis, abscesses, fistulas) and perianal disease were exclusive or almost exclusive features of CD. The occurrence of structural complications in CD was significant ($p<0.001$ for all) and constitutes the main differentiating feature (Table 3).

3.4. HOSPITALIZATION RATES

The variable "hospitalizations" refers to the total number of hospital admissions due to exacerbations or complications of IBD. The cumulative rate of hospitalizations due to IBD flares or complications was significantly different between the disease groups, serving as a key marker of disease severity and healthcare burden. The percentage of patients with CD who required hospitalization at least once (49.5%)

Table 4: Hospitalization rates			
Parameter	**Ulcerative Colitis (n=129)**	**Crohn's Disease (n=99)**	**IBD-U (n=16)**
Patients requiring ≥1 hospitalization (%)	27.1%	49.5%	25.0%
Mean number of hospitalizations (± SD)	0.5 ± 1.2	1.1 ± 1.9	0.4±0.7

was higher than that of patients with UC (27.1%) (p<0.001).

Table 5: Exposure to biologic and small-molecule therapies in the study cohort

Therapeutic Agent	Ulcerative Colitis (n=129)	Crohn's Disease (n=99)	IBD-U (n=16)	p-value
Mesalazine (Oral/Rectal)	114 (88.4%)	40 (40.4%)	11 (68.8%)	**<0.001***
Immunomodulators (AZA/MTX)	35 (27.1%)	63 (63.6%)	4 (25.0%)	**<0.001***
Biologic Agents (Any)	35 (27.1%)	55 (55.6%)	5 (31.3%)	**<0.001***
Anti-TNF Agents	22 (17.0%)	49 (49.5%)	4 (25.0%)	**<0.001***
Vedolizumab	6 (4.7%)	8 (8.1%)	0 (0.0%)	0.412
Ustekinumab	2 (1.5%)	5 (5.1%)	0 (0.0%)	0.284
Tofacitinib	5 (3.9%)	0 (0.0%)	0 (0.0%)	0.048*
Combination Therapy *(Biologic+Immunomodulator)*	15 (11.6%)	38 (38.4%)	2 (12.5%)	**<0.001***

The average number of hospitalizations per patient was also twice as high in CD (1.1) compared to UC (0.5) (Table 4), highlighting the higher burden of morbidity and intensive use of hospital resources associated with CD.

3.5. THERAPEUTIC MANAGEMENT

The utilization of various drug classes was assessed as cumulative exposure within the cohort, stratified by IBD type (Table 5). Concerning the individual drug classes, the data indicate that the use of Mesalazine was significantly higher in the UC group (88.4%) than in the CD (40.4%), reflecting its primary role in UC management (p<0.001). Although the numerical use of corticosteroids was higher in CD (60.6%) than in UC (52.7%), this difference did not reach statistical significance (p=0.087). Immunomodulator use (azathioprine or methotrexate) was significantly more frequent in CD (63.6%) compared to UC (27.1%) and IBD-U (25.0%) (p<0.001).

Exposure to biologic agents was also significantly higher in CD (55.6%, 55/99) than in UC (27.1%, 35/129) and IBD-U (31.3%) (p<0.001), largely driven by the greater use of anti-TNF agents (49.5% vs 17.0%, 22/129 and 25.0%, respectively; p<0.001). No significant differences were observed in the use of vedolizumab (p=0.412) or ustekinumab (p=0.284) across disease subtypes. Tofacitinib, was administered exclusively in patients with UC (3.9% of UC cohort). Notably, no cases of herpes zoster or thromboembolic events were recorded in this group. Combination therapy (biologic plus immunomodulator) was significantly more common in CD (38.4%) compared to UC (11.6%) and IBD-U (12.5%) (p<0.001). A small proportion of patients were exposed to more than one biologic agent during the course of their disease, mainly due to loss of response or intolerance to previous therapy.

3.6. SURGICAL OUTCOMES

The cumulative rate of IBD-related surgery was analyzed as a marker of severe, complicated, or refractory disease and is presented in Table 6. As it is shown, patients with CD exhibited markedly higher rates of surgery (32.3%) compared with those with UC (8.5%) and IBD-U (0.0%), (p<0.0001). This four-fold difference aligns with the known natural history of CD, characterized by stricturing and penetrating complications requiring surgical intervention, particularly in a cohort with longer disease duration.

Table 6: Cumulative rate of IBD-related surgery in the three groups of patients

Outcome	UC (n=129)	CD (n=99)	IBD-U (n=16)	p-value
History of any IBD-related surgery	11 (8.5%)	32 (32.3%)	0 (0.0%)	<0.0001*
Segmental Bowel Resection	0 (0.0%)	31 (31.3%)	0 (0.0%)	<0.0001*
Subtotal colectomy	6 (4.7%)	1 (1.0%)	0 (0.0%)	0.217
Ileoanal Pouch Anastomosis	5 (3.9%)	0 (0.0%)	0 (0.0%)	0.086

4. DISCUSSION

This 12-year retrospective, single-center study describes the clinical characteristics, phenotypic distribution, and therapeutic trajectories of 244 adult patients with IBD followed at a private specialized IBD unit in Athens, Greece. Data of this type complement existing literature derived predominantly from public academic centers and national registries [1-5] and provide descriptive information on a cohort managed in a private setting.

The cohort was characterized by a high proportion of complicated phenotypes. Among UC patients, 58.9% presented with extensive colitis (Montreal E3), while among CD patients, 52.5% exhibited stricturing (B2) or penetrating (B3) behavior and 26.3% had perianal involvement. These proportions are numerically higher than those reported in

unselected population-based cohorts such as the European EpiCom cohort [11, 12] and are more comparable to cohorts described in tertiary referral centers. In a single-center retrospective design without a public comparator, this observation cannot be interpreted as a difference between private and public care models, but most likely reflects referral bias. Patients with longer disease duration, refractory disease, or need for advanced therapies may be referred to specialized units regardless of sector.

The observed therapeutic patterns are consistent with contemporary Treat-to-Target (T2T) principles as defined by STRIDE-II [14]. STRIDE-II recommends clinical response and remission as immediate and intermediate targets, normalization of C-reactive protein and fecal calprotectin as intermediate targets, and endoscopic healing, restoration of quality of life and absence of disability as long-term targets [14]. In this framework, regular assessment with objective biomarkers and cross-sectional imaging, including intestinal ultrasound and MRI, has been recommended to guide dose optimization and sequencing of therapies [13]. The role of biomarkers in T2T monitoring has been further discussed by Lutz et al. [15]. In our cohort, cumulative exposure to advanced therapies was 41.0% overall. No comparative inference regarding superiority of private versus public settings can be made from these descriptive data.

Organization of care in dedicated IBD units has been evaluated through quality indicator and certification programs. The nationwide certification program reported by GETECCU has shown that certified/integrated IBD units are associated with more complete multidisciplinary team structure, better documentation, and more frequent use of recommended treatment protocols [26]. Similarly, a specialized inpatient IBD service has been associated with improvements in selected quality-of-care measures [19, 23]. These observations pertain to organizational structure of specialized care and not to the funding model (private vs. public). Therefore, our findings should be interpreted as descriptive of outcomes within a single specialized private unit, without implication of superiority over public systems.

Regarding private versus public comparisons in the literature, a single-center comparative analysis from a tertiary unit in Latin America reported that biologic use was approximately twice as high in the private system compared with the public system, while no statistically significant differences were found in rates of surgery or hospitalization between sectors in that study [16]. This nuance is important, as higher biologic use alone does not imply different surgical outcomes. In a separate analysis from New York City, Klein et al. reported socioeconomic differences among patients hospitalized in private versus public hospitals [21], underscoring that socioeconomic status and access may confound comparisons between sectors.

Data on variation between hospitals in IBD outcomes and costs have also been examined in the IBD Value study by van Linschoten et al. [22]. That study evaluated variation in outcomes, patient-reported experience measures, and costs among hospitals for patients treated with advanced therapies in the Netherlands, finding that most variation was attributable to patient-level characteristics (intraclass correlation coefficients typically 0-2% for outcomes) rather than to the treating hospital. The study was not designed to compare private versus public healthcare models as distinct sectors, and therefore does not support direct inferences about private versus public sector superiority [22].

Consistent with this literature, our surgical outcomes over 12 years (32.3% CD, 8.5% UC) and mean hospitalizations (1.1 CD, 0.5 UC) are within ranges previously reported from Greek and international cohorts [17-20]. Within the limits of a retrospective descriptive study without a comparator public cohort, these rates can only be described as observed in this private cohort and cannot be used to claim lower or superior rates relative to public or non-specialized centers.

The present study has strengths and limitations inherent to its design. First, its retrospective, single-center design is subject to documentation bias. Second, selection bias is present, as the population served by a private unit represents a specific demographic with access to private insurance or private payment, limiting generalizability to the entire Greek IBD population. Third, the study lacks a contemporaneous public comparator and statistical power for comparative inference; therefore no conclusions regarding superiority or modernization of the national healthcare landscape by private units can be drawn. The main strength is the detailed phenotypic and therapeutic characterization of a relatively large, well-characterized cohort followed for a long period in a specialized private setting. In conclusion, specialized private IBD units manage a patient population that in this cohort was characterized by high phenotypic complexity. The clinical outcomes, advanced therapy exposure, and surgical rates documented describe this single-center experience and are aligned with international standards as descriptive benchmarks. Future prospective, multi-center

studies integrating data from both public and private health sectors are warranted.

5. CONCLUSIONS

In this single-center retrospective cohort of 244 patients followed in a specialized private IBD unit, we observed a high proportion of extensive UC and complicated CD phenotypes, frequent exposure to advanced therapies (41.0% overall), and IBD-related surgery rates of 32.3% for CD and 8.5% for UC. These findings describe the clinical complexity and management patterns observed in this specific private setting. Future prospective, multi-center studies incorporating both public and private centers are needed to evaluate phenotypic distribution, treatment patterns, and outcomes across healthcare sectors in Greece.

6. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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